Facilitator Guide

Scientific Communication: Writing, Slide Design and Presentation Skills

Neudata Consulting Ltd · Version 1.0, November 2026

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Using this guide

This guide lets a trainer who did not write the course run it. It covers two standalone versions: a three-hour practical workshop for 12 to 24 people and a one-hour talk for any size of audience. Exact timings are in the run sheets. The decks carry speaker notes for every slide, with a suggested script, a question for the room, things to watch for, and the basis for each claim.

> Keep the promise modest. This is an introduction with guided practice. Participants will leave with four products and some useful habits, and not with mastery of scientific writing or public speaking.

Who it is for

Participants know their subject but may struggle to communicate it. They are researchers preparing conference presentations, students preparing thesis defences, and professionals in hospitals and other organisations presenting evidence or recommendations. Many will present in English as an additional language. Examples include health and research settings, and the CoolRoom case gives non-clinical participants a home. No prerequisite statistics are needed beyond the idea of a percentage and a confidence interval. Facilitators should be comfortable explaining those two ideas.

What you must not claim

- That rules such as one slide per minute, six words per line or a minimum font size are research findings. They are rules of thumb.
- That the assertion-evidence approach is proven. One experiment supports a bundle of changes, on a recorded talk, with students.
- That the rubric is a validated instrument. It is a course feedback tool, so do not use it to grade people.
- That any fictional number is real. CareCall and CoolRoom are invented.

Each slide's notes end with a Basis line. If a participant challenges a claim, read that line and refer to References_and_Source_Map.

Preparing

Two weeks before

- Confirm numbers, room and equipment. A flat room with movable chairs suits trios far better than fixed rows.
- Check the deck on the real projector if you can. Run the back-of-room test on slides 34 to 38.
- Ask for an assistant if the group is above 18.
- Send participants this note: "Please bring one result from your own work, or a paragraph you are writing, with identifiers removed. If you do not have one, we provide fictional cases. Do not bring confidential or identifiable data."

On the day

Item	Quantity	Used in
Workshop deck (PowerPoint, PDF as backup)	1	Whole session
Participant workbook, or printed Worksheets 1 to 7	1 each	Whole session
Worksheet 6 rubric and Worksheet 5 rehearsal sheet	3 spare per trio	Rehearsal

Item	Quantity	Used in
Question cards (Worksheet 7), cut up if you wish	1 set per trio	Q&A practice
Slide_Redesign_Exercise.pptx	Shared link	Module 4
Blank paper and pens	1 each	Storyboard, redesign
Visible countdown timer	1	All activities

Forming groups for 12 to 24 people

Pairs work through Modules 1 to 4. Trios work on the rehearsal and Q&A, formed by joining pairs at the break. Mix fields and, if possible, first languages, so that each listener has to rely on clear explanation.

Participants	Trios	Groups of four	Facilitators
12	4	0	1
13	3	1	1
14	2	2	1
15	5	0	1
16	4	1	1
17	3	2	1
18	6	0	1
19	5	1	1 plus assistant
20	4	2	1 plus assistant
21	7	0	1 plus assistant
22	6	1	1 plus assistant
23	5	2	1 plus assistant
24	8	0	1 plus assistant

A group of four does not get extra time, so it uses a shorter round: see the rehearsal section.

Three-hour workshop run sheet

The session totals exactly 180 minutes, including a 12-minute break. Slide numbers refer to the workshop deck. Clock times assume a 09:00 start.

Start	Elapsed	Min	Block	Slides	How the time is spent
09:00	0:00 to 0:08	8	Welcome and working pairs	1-4	Goals, ground rules, confidentiality, form pairs
09:08	0:08 to 0:28	20	Module 1: Audience, purpose and message	5-9	Teach 7, Worksheet 1 for 10, pair check 3
09:28	0:28 to 0:58	30	Module 2: Scientific writing	10-21	Teach 13, paragraph edit 12, debrief 5
09:58	0:58 to 1:18	20	Module 3: Presentation structure	22-30	Teach 8, storyboard 10, share 2
10:18	1:18 to 1:30	12	Break	31	Short break
10:30	1:30 to 2:04	34	Module 4: Slide design and data communication	32-51	Teach 16, critique and redesign 13, debrief 5

Start	Elapsed	Min	Block	Slides	How the time is spent
11:04	2:04 to 2:14	10	Module 5: Delivery skills	52-56	Teach and demonstrate 5, opening and transition drill 5
11:14	2:14 to 2:36	22	Timed rehearsal in trios	57-59	Brief 2, three 6-minute rounds, whole-group debrief 2
11:36	2:36 to 2:48	12	Question-and-answer practice	60-61	Teach and brief 3, three 3-minute rounds
11:48	2:48 to 2:55	7	Module 6: Accuracy and responsible use	62-66	Teach 4, accuracy check on own slide 3
11:55	2:55 to 3:00	5	Close: revise and commit	67-68	Choose one revision, write next step, course feedback
12:00	3:00	180	**End**		

Slides 69 to 71 are backups: the rubric at a glance, how the numbers were calculated, and evidence types. They carry no time allowance.

If you fall behind

Protect the rehearsal and Q&A. They are the reason to hold this workshop in person. Cut in this order.

If you are behind by	Do this	Saves
Up to 5 minutes	Merge slides 23 and 24, and take only one before-and-after pair in detail in Module 4.	3 to 5
6 to 10 minutes	Show only pairs 1, 3 and 4 in Module 4, and shorten Module 3 by dropping slide 26. Keep the defence slide if students are present.	6 to 8
More than 10 minutes	Drop the module-level pair share at the end of Module 3, and reduce Module 6 to the accuracy checklist.	4 to 5 more

Do not shorten the trio rounds. If you must lose a round, tell groups to run two rounds, and have the third speaker go first in the Q&A practice.

Block-by-block instructions

Welcome and working pairs (8 minutes, slides 1-4)

Purpose. Set expectations, make feedback safe and introduce the fictional cases.

Min	What happens
1	Title slide. Welcome. Timer visible.
3	Four products and the schedule bar. Take two answers to 'what would make today useful?'
2	Ground rules. Point out that accent and personality are never assessed, and ask people to remove identifiers.
2	Introduce CareCall and CoolRoom. Ask why called patients might differ. Pair people with a neighbour.
8	**Total**

Common problems

- Late arrivals. Pair them with each other rather than waiting.
- Someone has brought real patient data. Ask them to use the fictional case today.

Module 1: Audience, purpose and message (20 minutes, slides 5-9)

Purpose. Each person writes a one-sentence message for a real audience.

Min	What happens
7	Teach: section slide, audience chain, topic v message, three rooms.
10	Worksheet 1, alone. Circulate and ask 'what is the claim?'
3	Pair check. Partner says back what they think they are being asked to do.
20	**Total**

Debrief questions

- What did your partner think you wanted?
- What did you have to drop from your sentence?

Common problems

- Topics in place of claims. Ask for a verb.
- Audience of 'everyone'. Ask who matters most in the room.
- Messages stronger than the evidence. Ask what the data would say to that sentence.

Module 2: Scientific writing (30 minutes, slides 10-21)

Purpose. Each person revises one paragraph so that it says no more than the data allow.

Min	What happens
13	Teach: paragraph anatomy, three repairs, result v interpretation, denominators, size and uncertainty, verbs, non-significance, worked rewrite, four forms.
12	Worksheet 2. Alone for 8, swap and comment for 4.
5	Debrief. Two pairs read their opening sentence. Show the model answer.
30	**Total**

Debrief questions

- What did you delete first?
- Where did you want a stronger verb than the evidence allows?
- What would you need before claiming the coating works?

Common problems

- Time. This module is dense. If short, skip the sentence repairs slide, which is in the workbook.
- Participants worried by statistics. Say that you need the ideas of size and range, not the formula.
- Debates over p-values. Keep to the point: report the estimate and interval, and do not read p above 0.05 as no effect.

Module 3: Presentation structure (20 minutes, slides 22-30)

Purpose. Each person drafts a short storyboard built as an argument.

Min	What happens
8	Teach: one argument, openings, selecting methods, prioritising, three shapes, defences, endings.
10	Worksheet 3. Headlines on six cards, with a sketch of the evidence.
2	Read headlines to a partner: the title test.
20	**Total**

Debrief questions

- Do your headlines tell the story alone?
- Which card would you cut with half the time?

Common problems

- Students anxious about defence rules. Stress that formats vary, and that they should read their own regulations and ask their supervisor.
- Managers who think this is too academic. Point to the briefing shape, with the recommendation first.

Break (12 minutes, slides 31)

Purpose. Rest. Rearrange chairs and form trios.

Min	What happens
12	Break. Display the countdown. Check groups and rooms.
12	**Total**

Module 4: Slide design and data communication (34 minutes, slides 32-51)

Purpose. Each person redesigns one slide, with a claim and honest evidence.

Min	What happens
16	Teach: assertion-evidence and its limits, numeric rules, readability, chart choice, the five questions, misleading charts, five before-and-after pairs.
1	Introduce the exercise.
12	Critique in pairs for 3, redesign alone for 9. Leave slide 50 on screen.
5	Debrief. Two volunteers show sketches. Show the model.
34	**Total**

Debrief questions

- What did you remove first?
- What did you refuse to remove?
- What would someone at the back of the room still not see?

Common problems

- People polishing in PowerPoint. Paper is fine, and the headline comes first.
- Requests for 'the correct font size'. Explain the range and the room test. If pressed, say that 24 points for body text is a reasonable personal habit, not a finding.
- Removing uncertainty to simplify a slide. Ask whether the claim is still true without it.

Module 5: Delivery skills (10 minutes, slides 52-56)

Purpose. Each person says their opening and one transition aloud.

Min	What happens
5	Teach and demonstrate: four moves, clarity not accent, rehearsal and cutting.
5	Pair drill: opening and transition, 2 minutes each, plus 30 seconds to swap, and 30 seconds' spare.
10	**Total**

Common problems

- Reluctance to speak. Everyone speaks at once, which reduces exposure.
- Self-criticism about accent. Repeat that intelligibility and comprehensibility are the goals. Accent is not assessed.

Timed rehearsal in trios (22 minutes, slides 57-59)

Purpose. Each person gives a timed explanation and receives structured feedback.

Min	What happens
2	Brief: feedback prompts and the rubric. Form trios.
18	Three rounds of 6 minutes: speak 2, write 1, discuss 2, redo 1.
2	Whole-group debrief.
22	**Total**

Debrief questions

- Where did your two minutes go?
- What did a listener hear differently from what you meant?
- What will you cut first?

Common problems

- Rounds running over. Use one global timer and call each change loudly. Stopping on time is the skill.
- Feedback about confidence or accent. Redirect to the three prompts.
- Silent listeners. The write minute exists so that everyone has something to say.

Groups of four run four rounds of 4.5 minutes: speak 2, write 0.5, discuss 1.5, redo 0.5. That is 18 minutes, so they finish with the trios.

Question-and-answer practice (12 minutes, slides 60-61)

Purpose. Each person answers one difficult question and says what they do not know.

Min	What happens
3	Teach and brief: four steps, question types, one demonstration.
9	Three rounds of 3 minutes: question 0.5, answer 1, feedback 1.5.
12	**Total**

Common problems

- Long answers. One minute is the limit.
- Unfair questions. Keep it practice: fair, not hostile.

Groups of four run four rounds of 2.25 minutes: question 0.25, answer 1, feedback 1.

Module 6: Accuracy and responsible use (7 minutes, slides 62-66)

Purpose. Each person checks their slide for accuracy and responsible use.

Min	What happens
4	Teach: simplify the words and keep the claim, credit and confidentiality, AI.
3	Accuracy checklist on the participant's own slide.
7	**Total**

Common problems

- Requests for legal advice. Say that the rules differ by country and organisation, and refer people to their governance office.
- Strong views on AI. Keep to the verifiable points: humans are responsible, verify everything, keep confidential data out of unapproved tools.

Close: revise and commit (5 minutes, slides 67-68)

Purpose. Each person chooses one revision, one rehearsal and one check.

Min	What happens
5	Personal plan, course feedback form, one volunteer, thanks.
5	**Total**

Model answers

Worksheet 1: example for CareCall

Field	Hospital management
Occasion and limit	Quality committee, 10 minutes
Who	Medical director, finance lead, nursing lead. Busy. Mixed statistics knowledge.
What they know	Readmissions are a concern. They have not seen these data.
What they need	A decision on funding, with honest risks.
What should change	Approve a staggered trial.
Message	We recommend a six-month staggered trial on two wards before funding full roll-out, because the benefit could be 9.4 fewer to 0.8 more readmissions per 100.

Good answers for the other rooms:

- Conference: In two hospitals, a nurse call was linked to 6.9 fewer readmissions per 100, but much of that gap reflected who could be reached.

- Thesis committee: I show that reachability confounds observational evaluations of follow-up calls, and I defend the stratified analysis that reveals it.

Worksheet 2: model paragraph

> We asked whether a reflective roof coating lowers classroom temperature. Mean afternoon temperature was 29.4 °C in 12 coated classrooms and 30.6 °C in 12 standard classrooms, a difference of 1.2 °C (95% CI 2.5 °C cooler to 0.1 °C warmer). The interval includes no difference and a useful cooling, so this small pilot cannot settle the question. Head teachers chose the coated rooms, so other differences between rooms could explain some of the gap. We did not measure exam results.

The seven problems. Expect participants to find at least five.

Text in the exercise	Problem	Repair
"In light of the fact that", "constitutes", "the implementation of ... was undertaken"	Wordy opening. Nominalised verbs hide who did what.	Start with the question.
"significant problem"	"Significant" is used in two senses in one paragraph.	Keep the word for its statistical sense, or drop it.
"a number of classrooms"	Vague, with no denominator.	State 12 and 12.
"much lower"	No size or uncertainty.	Give the means, the difference and the interval.
"showing that the coating cools"	Causal claim from a non-random comparison.	Say what was compared and how rooms were chosen.
"will improve exam results"	Exam results were not measured.	Remove it, or state clearly that it was not studied.
"it can be concluded that the coating does not work"	Treats a non-significant result as proof of no effect. It also contradicts the earlier sentence.	Say the interval includes both no difference and useful cooling.

Accept other versions if they keep the means, the difference, the interval, the number of rooms and the way rooms were chosen, and if they make no claim about exam results.

Worksheet 3: model storyboard for CareCall

Card	Headline	Evidence
1 Problem	Nearly one in five patients came back within a month, so we asked whether a nurse call would help	154 of 850 patients (18.1%)
2 What we did	We compared 850 patients who did or did not receive a call within 72 hours	Flow diagram
3 Main finding	Readmissions were 6.9 per 100 lower among those called	Bars with denominators
4 Limit	The gap narrows to 4.3 per 100 within risk groups, and the interval crosses zero	Interval plot
5 Meaning	Calls may help, but reachable patients were healthier, so we cannot yet separate the two	Risk mix: 29% v 45% high risk
6 Conclusion	Test calls in a staggered trial before funding full roll-out	Decision box

Worksheet 4: critique and model redesign

Problems in the slide: a topic title; a table with no message; green and red as the only signal; 'NS' with no sizes or intervals; the claim that calls 'had no effect in subgroups'; and an unexplained contrast with the overall result. The model redesign is slide 51 of the workshop deck and pair 6 of Before_After_Slides.pptx.

> Model headline: In both risk groups fewer patients who got a call were readmitted, and the intervals are wide. Evidence: paired bars (26.7 v 29.8 per 100 at high risk, 9.6 v 14.6 at lower risk), with intervals for each difference (3.1 fewer, 13.3 fewer to 7.0 more; and 5.0 fewer, 10.6 fewer to 0.6 more).

Accept any honest alternative, such as a dot plot of the differences alone. Do not accept a headline that drops 'not randomised' from the talk altogether, or one that says calls worked in both groups.

Question cards: what a good answer sounds like

Type	Good answer
Critical	"That is a fair concern. Calls were not randomised, and patients reached were healthier. Stratifying by risk reduced the gap, which is why we recommend a randomised evaluation."
Unclear	"Do you mean the overall difference, or the difference within risk groups?" Then answer the one they mean.
Out of scope	"We did not study costs per readmission avoided. We have nurse time only. I can share that afterwards."
Unknown	"I do not know. We did not analyse patients over 85 separately. I can check whether the numbers allow it and reply by email."

One-hour talk run sheet

The talk stands alone. It needs no workbook, and it covers a realistic subset of the workshop: fewer slides, one audience activity, and no rehearsal. Do not try to rush through the workshop deck. The schedule totals exactly 60 minutes.

Start	Elapsed	Min	Block	Slides	How the time is spent
09:00	0:00 to 0:05	5	Opening: one finding, three rooms	1-3	Same fictional finding told three ways
09:05	0:05 to 0:15	10	Decide the audience and the message	4-6	Audience, purpose, one-sentence message
09:15	0:15 to 0:25	10	Claims that match the evidence	7-10	Effect sizes, denominators, uncertainty, causal language
09:25	0:25 to 0:32	7	Audience activity: fix the headline	11-12	Think 2, pairs 3, debrief 2
09:32	0:32 to 0:44	12	Slides and charts that explain	13-20	Headline plus evidence, three redesigns, misleading axis
09:44	0:44 to 0:51	7	Delivery, rehearsal and responsible use	21-24	Timing, explaining a figure, honest answers, AI checks
09:51	0:51 to 1:00	9	Questions and close	25-27	Questions 7, close 2
10:00	1:00	60	**End**		

What the talk leaves out

Paragraph editing, storyboarding, slide redesign, rehearsal, Q&A practice and detailed defence guidance are all workshop content. The talk mentions defences briefly under audience and says that requirements vary. It keeps the ideas with the highest value for a single hour: a one-sentence message, claims that match the evidence, a headline plus evidence, honest charts, timed rehearsal and honest answers.

Running the audience activity

Seven minutes: 2 alone, 3 with a neighbour, 2 for debrief. Works in fixed rows. With more than 100 people, take examples from a few rows or use a poll. Online, ask for headlines in chat, then read three aloud.

Handling questions

Seven minutes are reserved. Model the four steps as you answer: listen, repeat, answer briefly, check. If nobody asks, use the prompts on slide 25. Stop questions at minute 58 and close on slide 26.

Adaptations

Larger groups

Group size	Change	Rehearsal
25 to 40	Two facilitators. Tables of 6, holding two trios each. Worksheets as in the standard version.	Trios, standard timings. Facilitators sample tables and do not try to hear everyone.
41 to 60	Three facilitators or table leaders briefed in advance. Debriefs are sampled from three tables only.	Trios, standard timings. Print the round on each table and use a loud timer.
More than 60	Use the one-hour talk, or split into parallel workshops. If neither is possible, keep the schedule and run pairs.	Pairs: two rounds of 9 minutes (speak 2, write 1, discuss 3, redo 1, change 2). That is 18 minutes.

Online delivery

The online version keeps the 180-minute total. It uses the same deck. Assign trios in advance, and name breakout rooms. Ask for cameras on during rehearsal if people are comfortable, and allow a voice-only option. Tell participants to have the PDF workbook and a second window or a phone timer. Breakout moves cost time, so the online timings below take that from the briefs and debriefs and keep the trio rounds intact.

Block	Min	Online change
Welcome	8	Use chat for 'what would make today useful'. Assign pairs for the morning.
Module 1	20	Worksheet alone with cameras off. Pair check in two-person rooms: 3 minutes includes moving.
Module 2	30	Share the exercise paragraph in chat. Swap by chat or shared document.
Module 3	20	Storyboard in the worksheet or on paper. Read headlines in rooms.

Block	Min	Online change
Break	12	Leave a countdown on screen.
Module 4	34	Redesign in Slide_Redesign_Exercise.pptx, or on paper with a phone photo. Debrief by screen share.
Module 5	10	Drill in pair rooms: 4 minutes plus 1 minute to move.
Timed rehearsal	22	Brief 2, open rooms 1, three rounds of 6 minutes, rooms close and debrief 1. Use the broadcast message to call each change.
Q&A practice	12	Teach 2, open rooms 1, three rounds of 3 minutes. Stay in the same rooms if the platform allows.
Module 6	7	Accuracy check alone.
Close	5	Feedback by online form.
Total	**180**	

For more than 24 people online, add a co-host to manage rooms. Check before the session that participants can enter breakout rooms from their device. If the connection fails, send everyone the PDFs and restart with the next timed block.

Presenting in English as an additional language

- Send the workbook before the session, so that people can read it in advance.
- Speak at a steady pace. Pause after numbers, and write key terms on the board.
- Allow writing in another language first, then a move to English.
- Offer the phrases on Worksheet 7. Set phrases help when thinking time is short.
- Never correct accent. Correct a word only if the listener has misunderstood it, and do so privately.

Accessibility

- Offer materials in advance, and PDF or Word on request. The slide titles are unique, and charts carry alt text.
- Describe each figure aloud using the four moves. Do not refer to things by colour alone.
- Use a microphone where available, and repeat questions from the room.
- Offer a seated or written alternative for anyone who prefers not to rehearse aloud. They can act as timekeeper and give written feedback.

Alignment and feedback

Outcomes, activities and feedback

Outcome	Taught and practised in	Product	Rubric criteria
LO1. Define their audience, purpose and main message.	Module 1 and Worksheet 1	Audience-and-message worksheet, pair check	Message
LO2. Build an argument that connects a question, evidence, interpretation and implications.	Modules 2 and 3, storyboard	Revised paragraph and storyboard	Message, Structure, Evidence

Outcome	Taught and practised in	Product	Rubric criteria
LO3. Write clear scientific paragraphs, abstracts and slide headlines without overstating findings.	Module 2 exercise, Worksheet 2, Module 6 check	Revised paragraph, redesigned slide headline	Writing, Evidence
LO4. Design readable slides and explain figures accurately.	Module 4 exercise, Worksheet 4, trio rehearsal	Redesigned slide, spoken figure explanation	Visual clarity, Evidence
LO5. Deliver a timed presentation and respond to questions.	Module 5 drill, trio rehearsal, Q&A practice	Timed explanation, practised answers	Timing, Responses to questions
LO6. Use structured feedback to revise and rehearse their work.	Feedback prompts in rehearsal and Q&A, closing plan	Feedback notes, personal revision plan	All criteria

Using the rubric

Worksheet 6 is a feedback tool for practice, built for this course. It has not been validated. It focuses on seven criteria: message, structure, writing, evidence, visual clarity, timing and responses to questions. In the rehearsal, ask listeners to use only the rows that apply to what they heard. The ratings help a speaker choose one thing to change, so do not collect or total them. Nobody should comment on accent or personality.

Why the feedback is built this way

Hattie and Timperley's review found that information about the task and how to improve it has more value than comments on the person, and that feedback can also do harm. A meta-analysis by Double and colleagues found a modest average benefit from peer assessment, with a smaller estimate for university students. Most of that evidence comes from school or university study and not from talks, so we are extrapolating. The timed redo is our own teaching recommendation.

Fictional data sheet

Fictional teaching data. Not a real study. All figures below are generated by src/casebook.py, and scripts/check_numbers.py recalculates them independently. Use this sheet to answer questions.

CareCall (fictional). Two hospitals looked at whether a nurse telephone call within 72 hours of discharge was linked to fewer unplanned readmissions within 30 days. Of 1,037 adults discharged from medical wards in one year, 187 were excluded (62 transferred, 41 on palliative pathways, 84 with no working phone number), leaving 850. Calls were not randomised: patients received one when a nurse was free and the patient could be reached.

Group	Readmitted	Patients	Risk (%)
Call	60	412	14.6
No call	94	438	21.5
Call, high discharge risk	32	120	26.7
No call, high discharge risk	59	198	29.8
Call, lower discharge risk	28	292	9.6
No call, lower discharge risk	35	240	14.6

Quantity	Result
Crude risk difference, call minus no call	6.9 points lower (95% CI 1.8 to 12.0 lower)
Relative risk	0.68, a relative reduction of 32%
Within risk groups (Mantel-Haenszel)	4.3 points lower (95% CI 9.4 lower to 0.8 higher)
Emergency visits without admission	40/412 (9.7%) v 47/438 (10.7%); 1.0 lower (95% CI 5.1 lower to 3.0 higher)
Hospital A	95 of 540 (17.6%)
Hospital B	59 of 310 (19.0%)
Nurse time	412 calls at 20 minutes: about 137 hours. A simple illustration, not a costing.

CoolRoom (fictional). A school district painted reflective coating on the roofs above 12 classrooms and compared mean afternoon temperature over one hot term with 12 classrooms under standard roofs. Head teachers volunteered the coated rooms, so allocation was not random.

Facts you may use: 12 coated and 12 standard classrooms. Mean afternoon temperature 29.4 °C and 30.6 °C. Difference 1.2 °C cooler in coated rooms (95% CI 2.5 °C cooler to 0.1 °C warmer). Head teachers volunteered the coated rooms. Exam results were not measured.

Intervals use Wald standard errors for risk differences, the Greenland-Robins variance for the Mantel-Haenszel difference, and a t distribution with 22 degrees of freedom for the classroom difference.